INCH-POUND

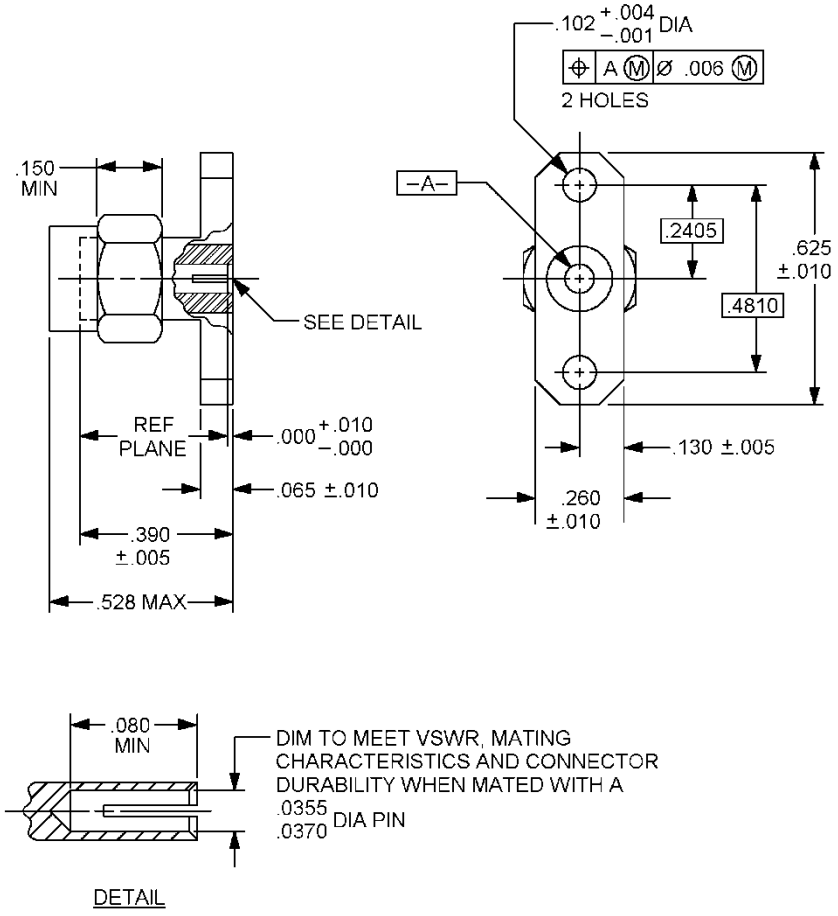
MIL-DTL-83517/11A
w/AMENDMENT 2
13 March 2025
SUPERSEDING
MIL-DTL-83517/11A
w/AMENDMENT 1
21 April 2015

DETAIL SPECIFICATION SHEET

# CONNECTOR, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, STRIP OR MICROSTRIP TRANSMISSION LINE, SERIES SMA (PIN CONTACT, FLANGE MOUNTED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83517.



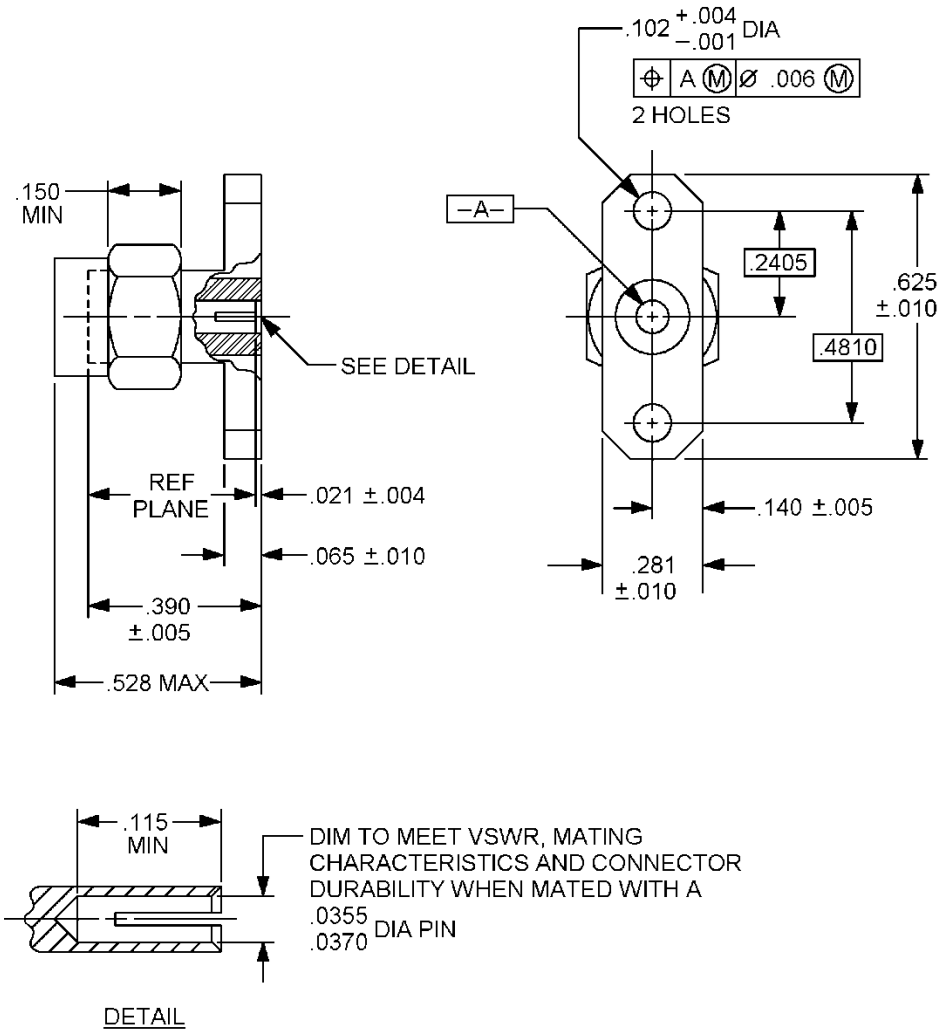

NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 1. Series SMA, pin contact, 2 hole (.260) flange mounted receptacle.

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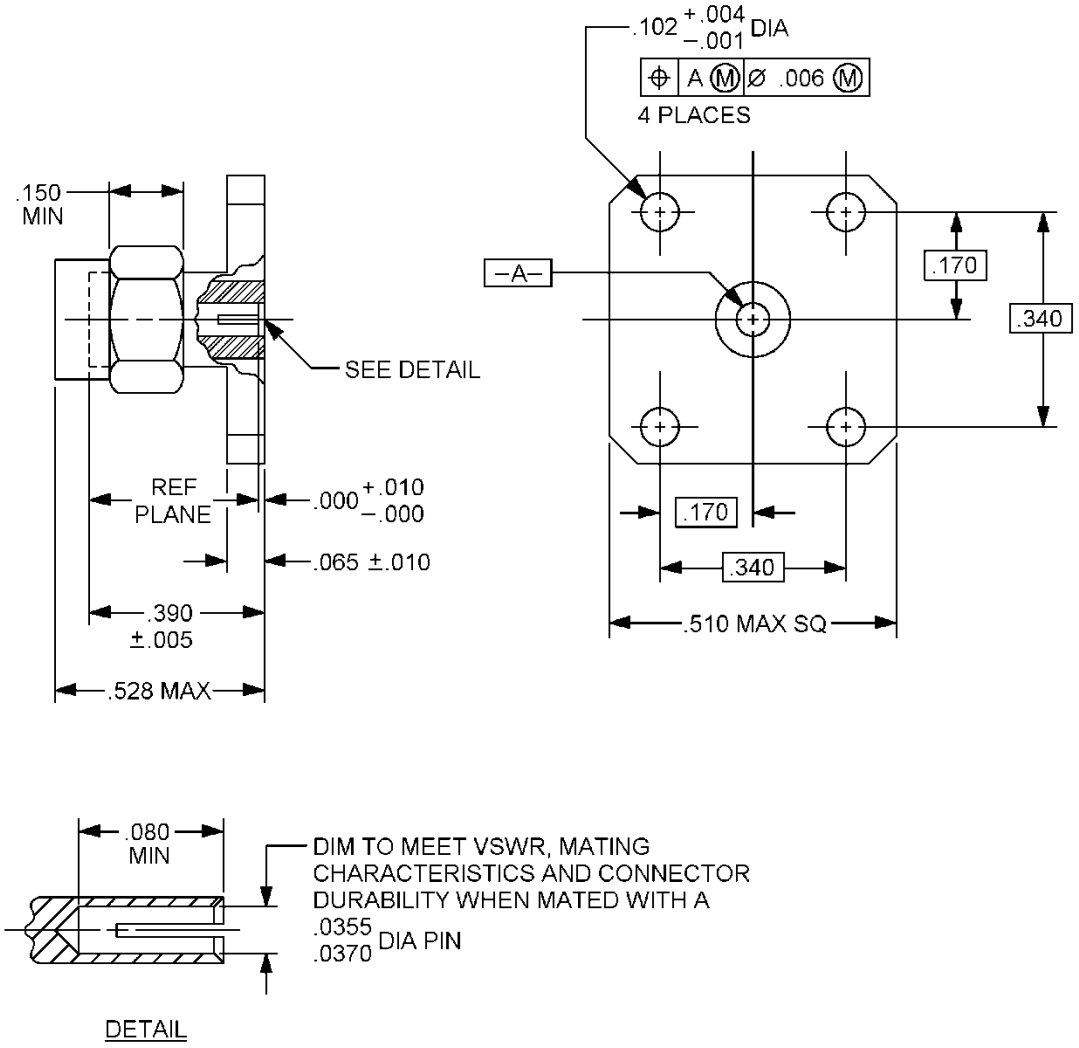



NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 2. Series SMA, pin contact, 2 hole, (.281) flange mounted receptacle.



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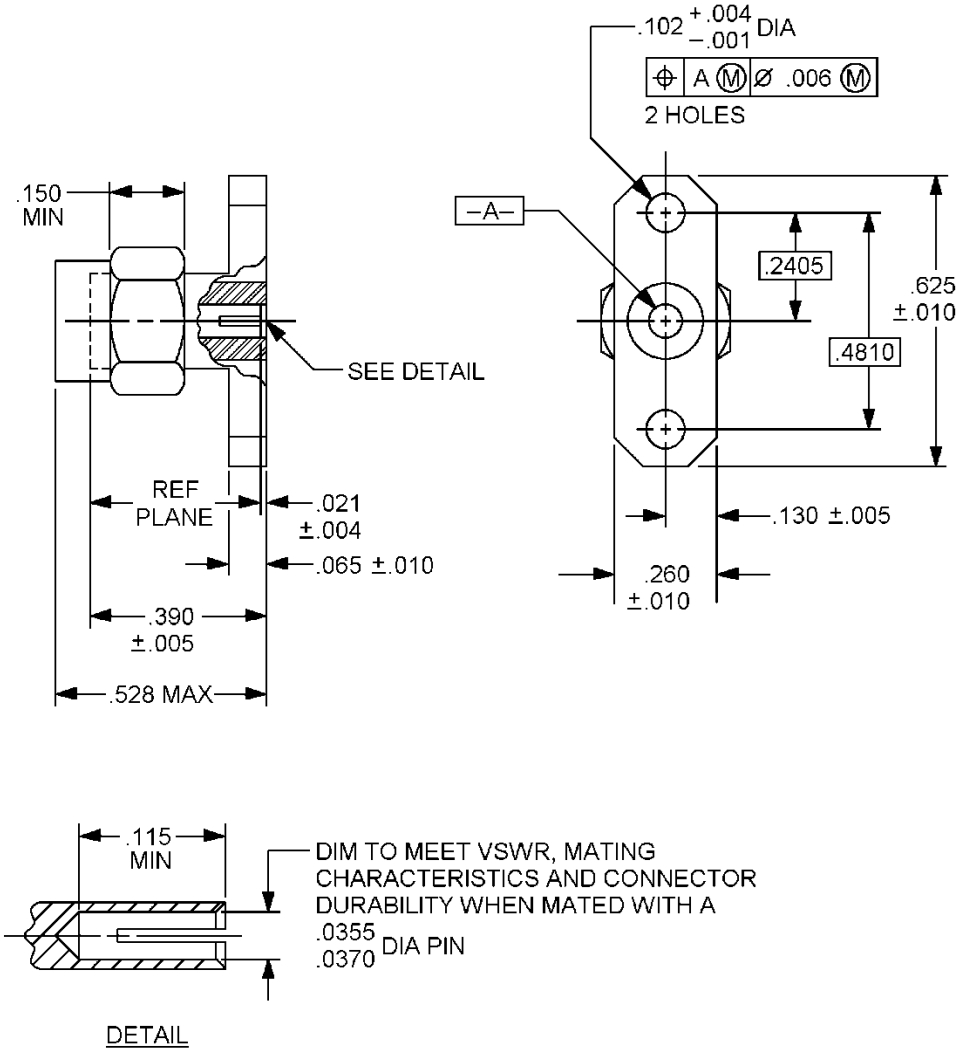

NOTES:

1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 3. Series SMA (.080 inch) pin contact, (4 hole) flange mounted receptacle.



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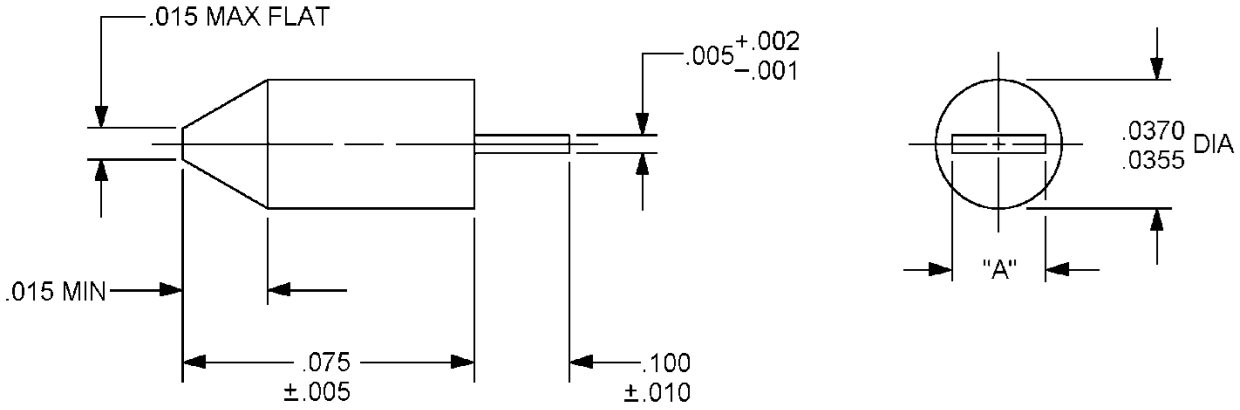



NOTES:
1. Dimensions are in inches.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 4. Series SMA (.115 inch) pin contact, (4 hole) flange mounted receptacle.



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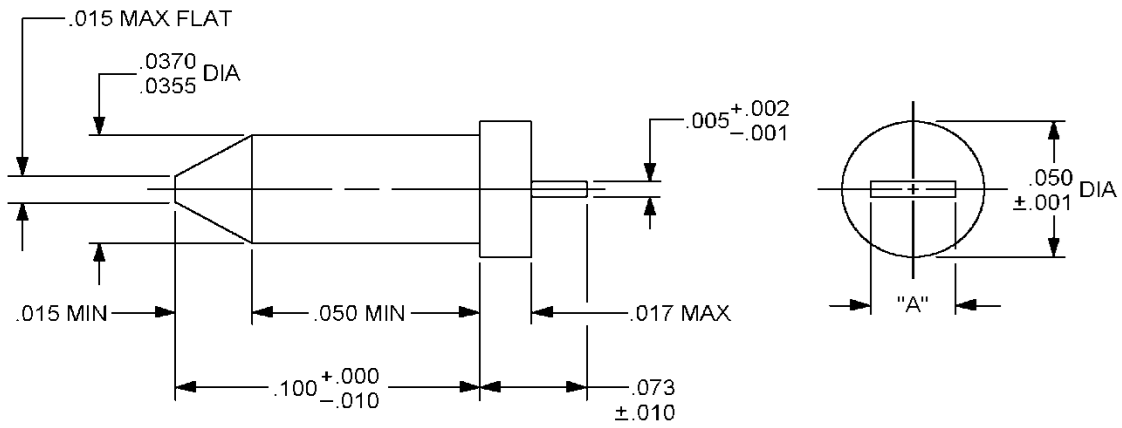

| "A" dim. |
|---|
| .020 ±.002 |
| .050 ±.002 |

NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 5. Transition pin (.075 inch).



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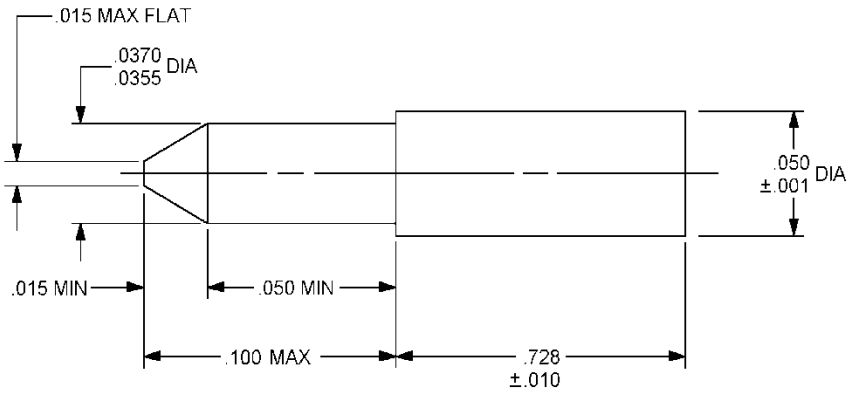

| "A" dim. |
|---|
| .020 ±.002 |
| .050 ±.002 |

NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 6. Transition pin (.100 inch).

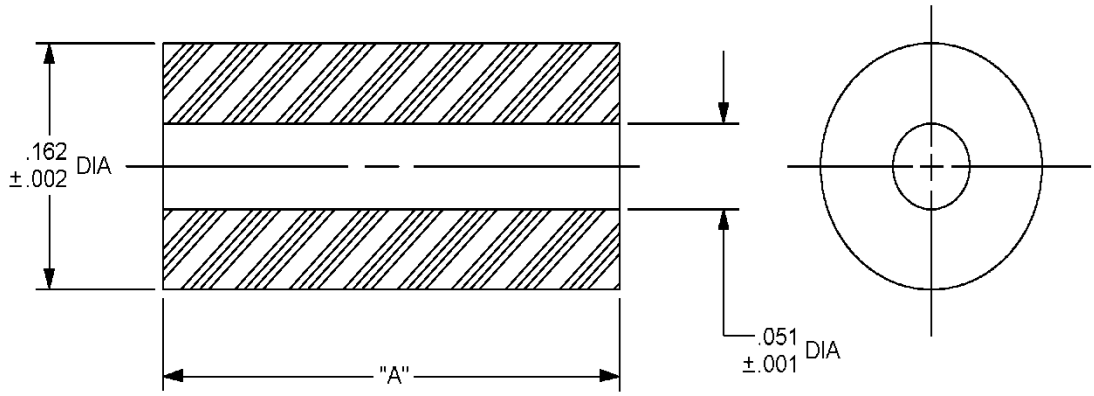


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NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 7. Transition pin.




| "A" dim. |
|---|
| .625 ±.002 |

NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Number two screws are suggested for mounting.

FIGURE 8. Rear dielectric.

ENGINEERING PARAMETERS

Nominal impedance: 50 ohms.

Voltage rating: 335 V rms maximum at sea level.
150 V rms maximum at 70,000 feet.



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Frequency range: 0 to 18.0 GHz.

Temperature rating: -65º to 105ºC.

REQUIREMENTS

Design and construction: See figures 1 through 8, table I and MIL-STD-348.

Force to engage and disengage:

Torque – 2 inch-pounds maximum.
Longitudinal force – Not applicable.

Coupling proof torque: 15 inch-pounds minimum.

Inspection note: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Contact gaging: See figure 9.

Contacts with spring members:

Center contact (socket):

Oversize test pin - .0375 +.0001.
Test pin finish – 16 microinches.
Insertion depth - .030/.045.
Number of insertions – 3.

Insertion force test: Steel test pin diameter .0370 + .0001.

Insertion depth - .050/.075.
Test pin finish – 16 microinches.
Insertion force – 3 pounds maximum.

Withdrawal force test: Steel test pin diameter .0355 - .0001.

Insertion depth - .050/.075.
Withdrawal force – 1 ounce minimum.
Test pin finish – 16 microinches.



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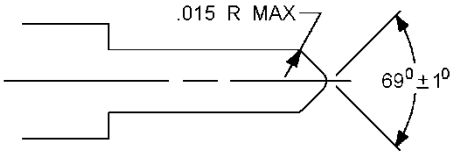



FIGURE 9. Test pin data.

TABLE I. Part or Identifying Number (PIN) and characteristics.

| PIN M83517/11- | Basic connector figure no. | Transition pin figure no. | Rear dielectric figure no. | Remarks |
|---|---|---|---|---|
| 31001 | 1 | - | - | - |
| 31002 | 2 | - | - | - |
| 31003 | 3 | - | - | - |
| 31004 | 4 | - | - | - |
| 31005 | 1 | 5 | - | .020 wide tab |
| 31006 | 1 | 5 | - | .050 wide tab |
| 31007 | 2 | 6 | - | .020 wide tab |
| 31008 | 2 | 6 | - | .050 wide tab |
| 31009 | 2 | 7 | 8 | .050 dia. terminal |
| 31010 | 3 | 5 | - | .020 wide tab |
| 31011 | 3 | 5 | - | .050 wide tab |
| 31012 | 4 | 6 | - | .020 wide tab |
| 31013 | 4 | 6 | - | .050 wide tab |
| 31014 | 4 | 7 | 8 | .050 dia. terminal |

Permeability of nonmagnetic materials: Applicable.

Seal:

Hermetic sealed connectors: Not applicable.
Pressurized and weatherproof connectors: Not applicable.

Insulation resistance: 5,000 megohms minimum, in accordance with test procedure EIA-364-21.

Center contact retention:

Axial force: 6 pounds minimum.
Torque: Not applicable.

Dielectric withstanding voltage: In accordance with test procedure EIA-364-20, test condition I.

Test voltage 1,000 V rms.

Corrosion: In accordance with test procedure EIA-364-26, test condition B.

Voltage standing wave ratio (VSWR): 1.35 +.01F (F in GHz), maximum.

Test frequency range: From .5 to 18.0 GHz. (Using a test method approved by the qualifying activity.)



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RF transmission loss: $.07\sqrt{F \text{ (F in GHz)}}$ tested from 2-18 GHz.

RF leakage: Not applicable.

Connector durability:

Interface:

500 cycles minimum at 12 cycles/minute maximum rate.
Connector shall meet contact gaging and force to engage and disengage requirements.

Transition pin:

500 cycles minimum at 12 cycles/minute maximum rate.
Contact shall meet contact gaging requirements.

Contact resistance: In milliohms maximum.

| | Initial | After environment |
|---|---|---|
| Center contact: | 6.0 | 8.0 |
| Outer contact: | 2.0 | Not applicable |

Thermal shock: Applicable, test condition A.

Moisture resistance: MIL-STD-202-106.

No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

RF high potential withstanding voltage:

At a frequency between 5 to 7.5 MHz.
Leakage current – Not applicable.
RF voltage – 670 V rms.

Coupling mechanism retention force: 60 pounds minimum.

PIN: M83517/11-(dash number from table I.)

Group qualification: See table II.



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TABLE II. Group qualification.

| Group | Submission and qualification of any of the following connectors 1/ 2/ | Qualifies the following connectors |
|---|---|---|
| 1 | 3+005 | 3+001<br>3+003<br>3+005<br>3+006<br>3+010<br>3+011 |
| II | 3+007 | 3+002<br>3+004<br>3+007<br>3+008<br>3+012<br>3+013 |
| III | 3+009 | 3+002<br>3+004<br>3+009<br>3+014 |

1/ Individual connectors other than listed are self qualifying only.
2/ Qualification of connectors, qualifies connectors of the same material only.
+ Denotes finish.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-DTL-83517, this document references the following:

MIL-STD-202-106
MIL-STD-348
EIA-364-20
EIA-364-21
EIA-364-26

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5935-2025-013)

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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